



Speed, Velocity & Acceleration

What
is Scalar?

A scalar is a physical quantity that has only a magnitude.

Examples:

- Mass
 - Length
 - Time
 - Temperature
 - Volume
 - Density
-



Length of a car is 4.5 m

physical quantity

magnitude



Mass of gold bar is 1 kg

physical quantity

magnitude



Time is 12.76 s

physical quantity

magnitude



Temperature is 36.8 °C

physical quantity

magnitude

What
is Vector?

A vector is a physical quantity that has both a magnitude and a direction.

Examples:

- Position
- Displacement
- Velocity
- Acceleration
- Momentum
- Force



The car is driving **north** at **98 km/h**

↗
direction

↖
magnitude



The rook moves 3 spaces to the left

magnitude



direction



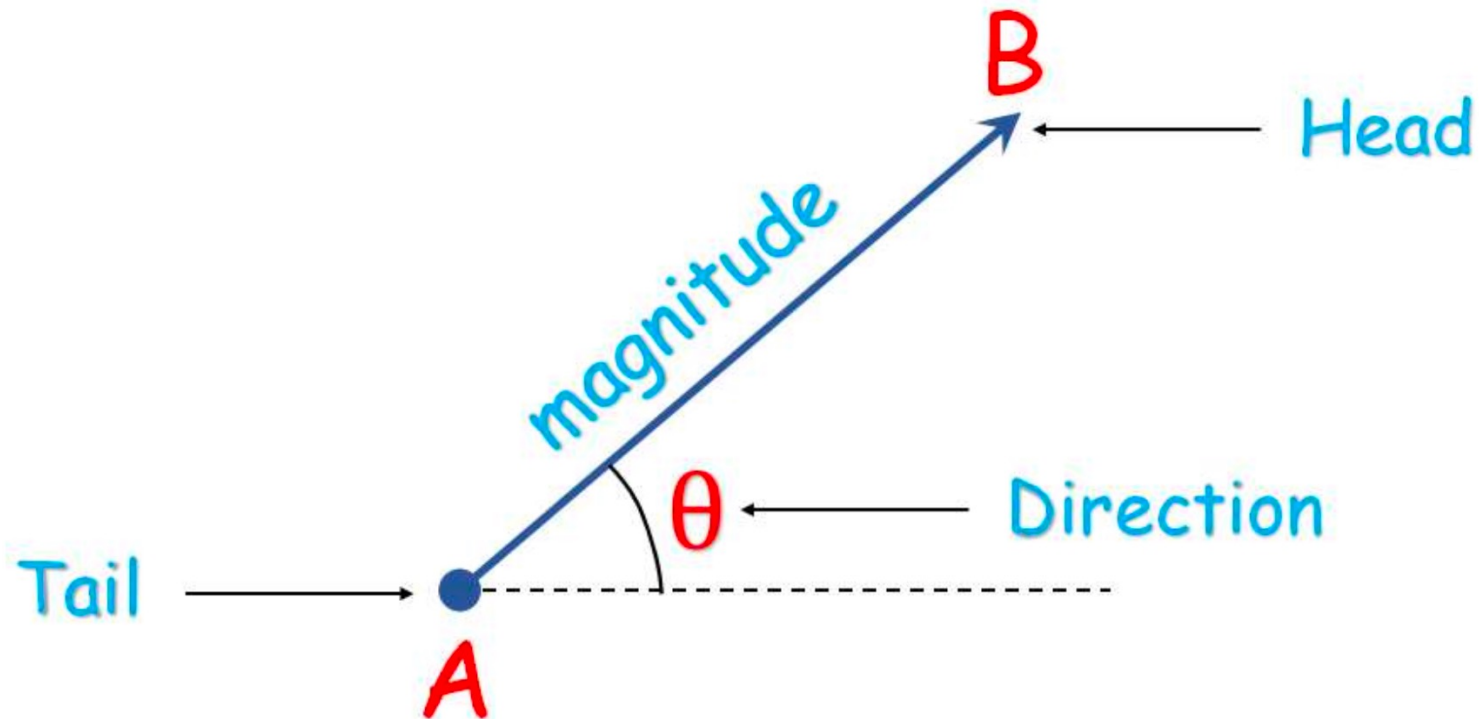


The snowboarder is accelerating **downward** at **0.5 m/s²**

direction

magnitude

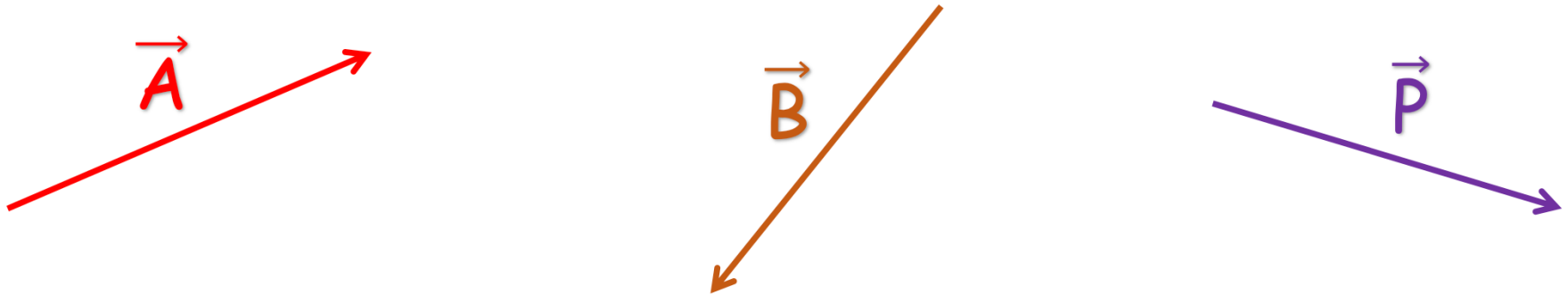
Representation of a vector



Symbolically it is represented as $\overrightarrow{AB}$

Representation of a vector

They are also represented by a single capital letter with an arrow above it.





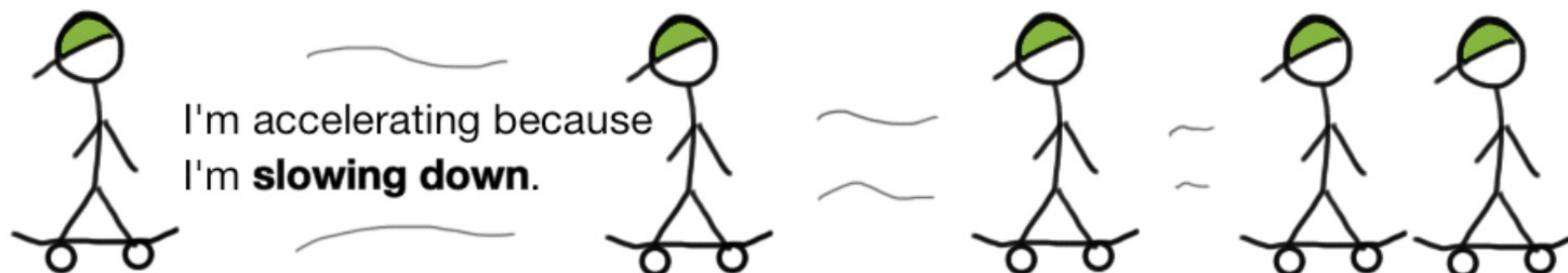
- 20°C

scalar

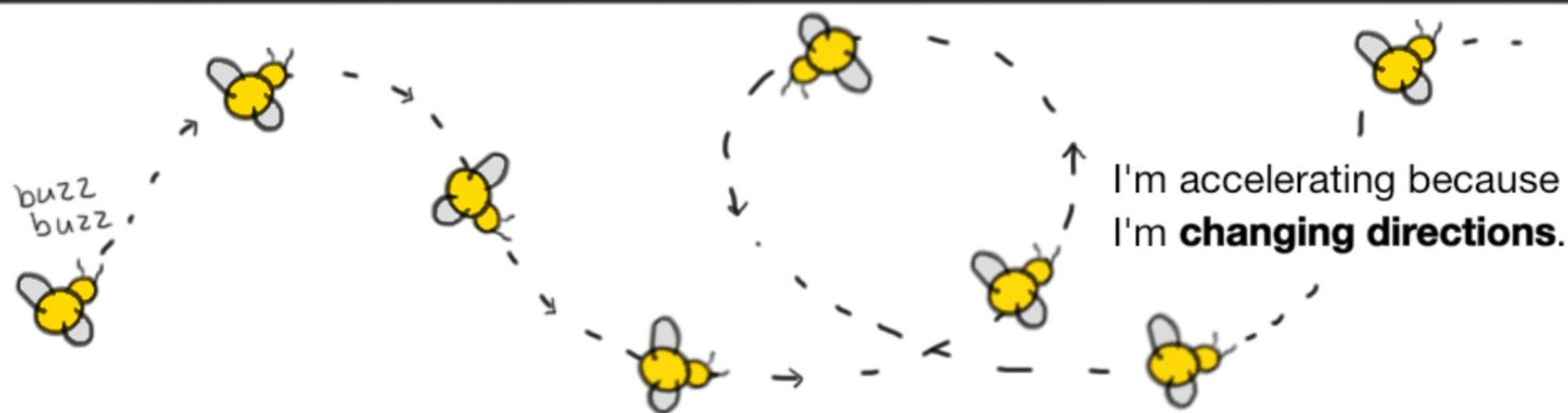
Quantity	Definition	Example	Formula
Speed	Scalar quantity showing how fast an object is moving	The car is moving at 11 m/s	$S = d/t$ $\text{Speed} = \frac{\text{distance}}{\text{time}}$
Velocity	Vector quantity showing how fast an object is moving in a particular direction.	The car is moving 11 m/s east down Fengqi Rd.	$v = d / t$ or $v = \frac{\Delta x}{\Delta t}$ $v = \frac{\text{change in position}}{\text{change in time}}$
Acceleration	Vector quantity showing the change in velocity	The car came to a stop in 1 second. -11 m/s²	$a = \frac{\Delta v}{\Delta t}$ $a = \frac{\text{change in velocity}}{\text{change in time}}$ $a = \frac{v_f - v_0}{t_f - t_0}$



I'm accelerating because
I'm **speeding up**.



I'm accelerating because
I'm **slowing down**.



I'm accelerating because
I'm **changing directions**.

Speed vs. Acceleration

Sort the following examples on a quadrant plane:

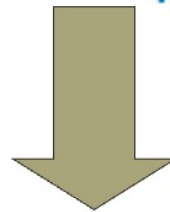
- A.) a rabbit sees a predator and starts to run full speed.
- B.) a rabbit runs full speed for as long as it can.
- C.) a rabbit running full speed changes direction to confuse the predator.
- D.) a rabbit is grazing peacefully in a field

High speed, low acceleration B.)	High speed, high acceleration C.)
low speed, low acceleration D.)	low speed, high acceleration A.)

Balanced Versus Unbalanced



$$\begin{array}{c} \text{---} \rightarrow + \leftarrow \text{---} = 0 \\ \text{Net Force} = 0 \end{array}$$



Balanced forces cause no acceleration.

Balanced Versus Unbalanced



$$\begin{array}{c} \rightarrow + \leftarrow = \rightarrow \\ \text{Net Force} = \rightarrow \end{array}$$



$$\begin{array}{c} \rightarrow + \rightarrow = \rightarrow \\ \text{Net Force} = \rightarrow \end{array}$$



Unbalanced forces
cause acceleration.